

NetSynergy: Mitigating Application-layer DDoS via Adaptive Access-Backbone Collaboration

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Abstract

Existing Distributed Denial of Service (DDoS) defense systems typically rely on computationally intensive mitigation mechanisms deployed within the victim-side access network to handle application-layer attacks. However, these systems prioritize accuracy over throughput and are therefore easily overwhelmed by high-volume attacks from the backbone networks. To address this limitation, we propose NetSynergy, an access-backbone collaborative DDoS mitigation system that enables high-throughput and accurate application-layer DDoS mitigation. NetSynergy adopts a collaboration-based mitigation model approach where backbone-side mitigation models are generated based on fine-grained access-side traffic analysis capabilities. These models are iteratively updated using a legitimate-traffic-preserving training strategy, and an adaptive model confidence controller is deployed to balance mitigation throughput and accuracy according to real-time defense status. Additionally, the traffic dynamics adaptation component in NetSynergy selects high-value traffic samples directly on the dataplane and accelerates model adaptation via the model update fast path mechanism. Evaluation on a hardware prototype demonstrates that NetSynergy effectively protects victim services under high-volume DDoS attacks, successfully forwarding up to 37.08% more benign traffic to the victim network compared with state-of-the-art dataplane traffic classification schemes.

CCS Concepts

• Security and privacy → Denial-of-service attacks.

Keywords

DDoS Mitigation, Programmable Data Plane, Decision Tree

ACM Reference Format:

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1 Introduction

Application-layer Distributed Denial of Service (DDoS) attacks have grown increasingly prevalent and high-volume in recent years [1, 7, 31, 40], posing substantial challenges to existing DDoS mitigation mechanisms. Current DDoS defense systems typically employ a static, attack-type-based division of labor [8, 22, 33]: backbone networks are tasked with filtering easily identifiable volumetric DDoS floods, while sophisticated application-layer DDoS attacks are relegated entirely to access-side systems. This architecture is becoming increasingly ineffective when defending against high-volume application-layer attacks.

The core limitation of existing DDoS defense systems lies in a widening gap between backbone visibility and access-side capacity. At the backbone, due to traffic encryption, user privacy concerns, and the potentially high cost of infrastructure upgrades, backbone systems typically rely on coarse-grained L3/L4 traffic features and simplified mitigation models, which can only filter prominent volumetric DDoS attacks [8, 11, 22, 23, 33, 50, 56]. Conversely, access-side mitigation systems rely on computationally intensive mechanisms, such as traffic decryption and packet payload inspection, to achieve accurate detection. However, their capacity is inherently constrained and cannot be cost-effectively scaled to handle high-volume attacks. Moreover, the capacity of access-side defense is limited by the bandwidth of the access-backbone link [8, 10, 22, 33, 53]. As a result, increasingly high-volume application-layer attacks [1, 7, 31, 40] easily evade backbone-side mitigation and flood the bandwidth-constrained access side, saturating the victim network's bandwidth and disrupting legitimate services.

Fortunately, the strengths of access-side and backbone-side mitigation systems are complementary, offering the potential for collaborative defense against high-volume application-layer attacks. While backbone-side systems can handle large-scale attacks, their fine-grained traffic analysis capabilities are inherently limited. In contrast, access-side systems, which process only traffic destined for their local networks and operate with constrained throughput, excel at fine-grained analysis. They can leverage richer traffic characteristics and more sophisticated models, such as packet payload fingerprinting [6, 28] and advanced machine learning algorithms [12, 30, 42, 43], to accurately identify malicious behavior. Therefore, when facing high-volume application-layer attacks, integrating the backbone-side high-throughput capacity with the access-side's fine-grained traffic analysis strengths presents a promising solution. By offloading a portion of the application-layer

DDoS mitigation workload to backbone-side systems, this access-backbone collaborative approach could enable an effective and scalable DDoS defense.

However, implementing an access-backbone collaborative defense raises two major challenges. Firstly, the backbone-side mitigation system offers limited capability in terms of available traffic features and mitigation models. Therefore, the backbone-side mitigation accuracy inevitably lags behind that of the access-side systems. Consequently, collaborative defense systems face a fundamental trade-off between mitigation throughput and accuracy. The system must therefore maximize accuracy while also dynamically adjusting its strategy to maintain sufficient overall mitigation throughput for service protection. Secondly, as both attack and legitimate traffic evolve continuously, the defense system must monitor traffic changes in real time and rapidly adapt its mitigation model accordingly. Moreover, partial traffic visibility in the access network, caused by upstream backbone-side mitigation, forces the system to build a dataplane-deployable monitoring mechanism.

To address these challenges, we propose NetSynergy, an access-backbone collaborative DDoS mitigation system. NetSynergy constructs backbone-deployable decision-tree-based mitigation models using the access-side detection results, and applies them to the backbone side for high-volume mitigation. Firstly, we design a decision-tree-based mitigation model generation mechanism. By strictly limiting the decision tree nodes to the feature space of the known dataset during the training process, we minimize the backbone-side false positive rate for benign traffic and ensure high overall mitigation accuracy. We further introduce an adaptive model confidence controller that monitors the system's mitigation status in real time, dynamically balancing the overall accuracy and throughput. Secondly, NetSynergy's traffic dynamics adaptation component incorporates a model update fast path mechanism to handle evolving traffic. This component prioritizes flows that exhibit high classification uncertainty or unstable traffic behavior in subsequent model updates, allowing the system to quickly adapt to emerging attack strategies or changes in legitimate traffic patterns. These prioritized samples are selected entirely on the dataplane at line rate, enabling backbone-side discovery of traffic dynamics.

We implement a prototype of NetSynergy using a Tofino hardware switch and DPDK-enabled servers, and evaluate its performance by replaying representative DDoS datasets in a testbed. Experimental results demonstrate that NetSynergy outperforms existing DDoS mitigation schemes, successfully forwarding up to 37.08% more benign traffic under high-volume DDoS attacks. Under dynamic attack scenarios, NetSynergy achieves robust mitigation, protecting 91.99% of benign traffic. Furthermore, we conduct an ablation study on NetSynergy, validate its line-rate mitigation capability, and assess the deployment overhead.

To sum up, we make the following contributions:

- We propose an access-backbone collaborative DDoS mitigation architecture that enables high-throughput and accurate attack mitigation. It leverages the complementary strengths of access-side and backbone-side systems through efficient mitigation model collaboration.
- We design an adaptive mitigation model generation component, which balances overall mitigation throughput and

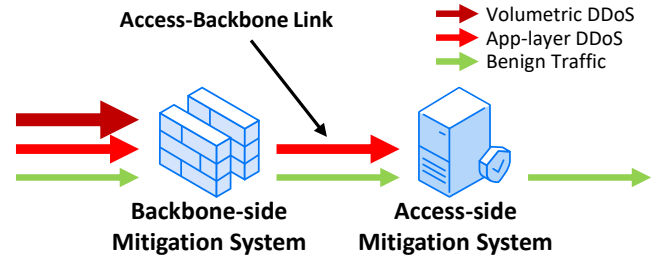


Figure 1: Current DDoS Defense Systems

accuracy with legitimate-traffic-preserving model generation and runtime model confidence control mechanism.

- We design a traffic dynamics adaptation component that prioritizes traffic with high model uncertainty or unstable traffic characteristics in model updates, enabling the system to rapidly adapt to evolving traffic patterns.

2 Background

2.1 Limitations of Current DDoS Defense

Existing DDoS mitigation schemes are generally categorized into two types: backbone-side (on-demand or cloud defense) and access-side (on-premises or edge defense) mitigation systems [8, 10, 22, 33, 53]. For backbone-side schemes, Internet Service Providers (ISPs) and Cloud Security Service Providers (CSSPs) deploy dedicated DDoS scrubbing centers to mitigate attacks in a centralized manner, where malicious traffic is diverted and filtered before reaching the target network [8, 10, 22, 33, 53]. Leveraging their bandwidth advantages, backbone-side schemes effectively mitigate volumetric DDoS attacks. However, they typically process only simple attacks and cannot guarantee that the processed traffic is attack-free [8, 22, 33]. For access-side schemes, victim networks often deploy their own DDoS mitigation systems, which not only address basic DDoS using on-premises devices [5, 27] but also employ intrusion detection systems (IDS) [6], web application firewalls (WAF) [28], and other security mechanisms to handle application-layer attacks. As illustrated in Figure 1, current mitigation architectures adopt a static, attack-type-based division of labor: backbone systems handle volumetric DDoS attacks, while access systems address application-layer DDoS attacks [22, 33].

The growing prevalence of application-layer attacks and traffic encryption has introduced new challenges to this defense architecture. In recent years, the scale of application-layer and encrypted attacks has continued to rise [1, 7, 31, 40]. These attacks cannot be effectively mitigated by existing backbone-side mitigation systems. For example, when targeting encrypted services, backbone-side systems typically cannot decrypt and inspect payloads, rendering payload-based classification methods unusable [6, 19, 28, 54]. Other coarse-grained approaches, such as performing filtering or rate limiting based on traffic statistics [2, 11, 23, 50], struggle to achieve high mitigation accuracy against such attacks. Therefore, existing DDoS defense systems increasingly rely on access-side systems to handle application-layer attacks and their growing volume.

However, the processing throughput of access-side mitigation systems is inherently constrained and cannot scale with rapidly increasing attack volumes. Firstly, as shown in Figure 1, the processing capacity of access-side defense is limited by the bandwidth of the access-backbone link [8, 10, 22, 33, 53]. An application-layer attack that exceeds this bandwidth can easily saturate the link and disrupt benign services. Secondly, the processing throughput of the access-side mitigation itself is also a bottleneck. Fine-grained access-side analysis relies on computationally intensive mechanisms such as traffic decryption, packet payload fingerprinting [6, 28], and advanced machine learning algorithms [12, 30, 42, 43]. None of these can be cost-effectively scaled to handle high-volume attacks.

In summary, high-volume application-layer attacks have become increasingly challenging for existing defense systems to mitigate. Because backbone-side mitigation cannot accurately identify and filter these attacks, malicious flows can easily bypass backbone defenses and overwhelm the victim network.

2.2 Opportunity

Access-Backbone Collaboration. Access-side and backbone-side mitigation systems have complementary strengths, offering a promising opportunity for enabling effective mitigation with both high throughput and high accuracy. Existing access-backbone collaboration schemes primarily focus on enabling automated coordination between the two separate systems [13, 25, 32, 40]. In these approaches, the access-side systems can request backbone-side assistance upon detecting an attack, significantly reducing backbone-side mitigation response time and increasing overall mitigation throughput. Some schemes further allow the access side to obtain backbone-side traffic monitoring information and specify mitigation actions, such as deploying access control list (ACL) or selecting predefined policies [25, 32], thereby improving flexibility. However, existing schemes do not enhance the backbone-side analysis, leaving it incapable of assisting in scrubbing application-layer attacks.

NetSynergy takes a step further in access-backbone collaboration: we enhance the backbone-side capabilities to support application-layer attack mitigation. Specifically, we enhance the backbone-side's fine-grained analysis capabilities, enabling it to execute flexible mitigation actions rather than merely utilizing existing predefined policies. This allows the adoption of more sophisticated mitigation policies, extending the benefits of access-backbone collaboration to application-layer attack scenarios and enabling effective defense against high-volume application-layer attacks.

Programmable Dataplane. Programmable dataplanes offer a new opportunity to enable intelligent defense on the backbone-side system. Programmable dataplane devices such as Tofino [16] can be reprogrammed using P4 [41] to support line-rate processing of user-defined functions. In an access-backbone collaborative defense system, the backbone-side system requires flexibility, enabling it to update the deployed mitigation strategy when improved alternatives are provided with the assistance of the access side. At the same time, it must also deliver sufficient processing throughput to meet the high-throughput requirements of the backbone networks. Programmable dataplanes offer a promising solution that satisfies both requirements. By leveraging the flexibility and high-throughput

capabilities of programmable dataplanes, we can implement fine-grained mitigation on the backbone-side system, forming a solid foundation for collaborative defense.

2.3 Threat Model

NetSynergy targets DDoS mitigation for privacy-sensitive victims such as government, medical, and military systems [21, 22, 38]. These users, who prefer to minimize third-party exposure of their traffic, are sensitive to CDN, Cloud WAF, and other L7 DDoS mitigation solutions that terminate TLS at the edge and expose application-layer information before traffic reaches the victim network. For such a deployment scenario, the backbone-side defense systems typically lack L7 inspection capabilities.

Under this problem scope, NetSynergy focuses on mitigating high-volume DDoS attacks, particularly application-layer attacks that are challenging for backbone-side mitigation systems to handle. Note that NetSynergy mainly focuses on bandwidth-consuming application-layer attacks, rather than computation-consuming ones that do not inflate traffic volumes [20, 44, 48]. The latter rely less on backbone-side assistance, as no access-backbone link bottleneck exists. NetSynergy targets the scenario where the attack traffic volume exceeds the access-backbone link capacity, which is also the typical situation in which access-side mitigation fails [8, 10, 22, 33, 53]. In this situation, assistance from the upstream backbone-side mitigation system is necessary; otherwise, legitimate traffic would be dropped before reaching the access network. Additionally, we assume that the backbone-side mitigation systems cannot accurately mitigate all attack traffic with existing policies, as backbone-side systems mainly focus only on volumetric DDoS [8, 22, 33]. For example, application-layer attacks or those targeting encrypted applications may bypass backbone-side mitigation.

We deploy NetSynergy as an access-backbone collaborative DDoS mitigation system. Both the access-side and backbone-side systems participate in attack mitigation, with their respective mitigation capacities consistent with the assumptions above. Traffic destined for the victim network first passes through the backbone-side mitigation system before being forwarded to the access-side system. Only traffic that passes through both mitigation systems finally reaches the protected services. Initially, the backbone-side system is not capable of mitigating the targeted attacks, while the access-side system can identify them accurately. As NetSynergy executes, the backbone-side system gradually learns to identify and mitigate the targeted attack traffic through access-backbone collaboration. Finally, the system achieves effective mitigation against high-volume DDoS attacks, meeting both throughput and accuracy requirements.

3 NetSynergy Overview

In this section, we first describe the general design guidelines for an access-backbone collaborative DDoS mitigation system, and then present the architecture of NetSynergy.

3.1 Design Guideline

The core challenge in designing an access-backbone collaborative defense system is how to enable backbone-side fine-grained DDoS mitigation using access-side insights, without incurring excessive

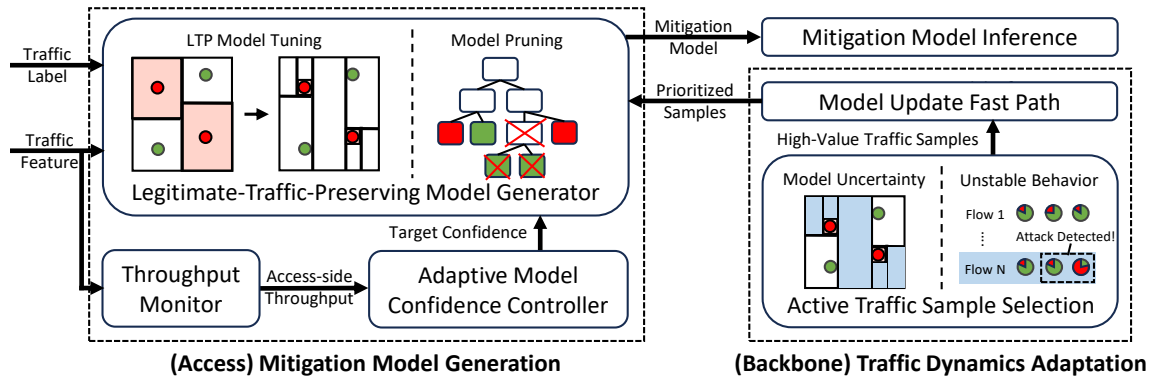


Figure 2: NetSynergy Architecture

cost or complexity. The most intuitive approach is to directly move or replicate the access-side application-aware defense upstream in the backbone network. However, implementing full access-side policies requires resource-intensive mechanisms such as traffic decryption and packet payload inspection. Scaling these mechanisms to meet backbone-side bandwidth requirements would incur prohibitively high costs, even though this approach would theoretically achieve optimal mitigation. In fact, this approach is similar to the existing CDN or Cloud WAF defense architectures and incurs relatively high costs to the customers [26].

Another approach is to directly apply access-side analysis results in the backbone network. For example, the access-side system can transmit malicious flow identifiers to the backbone-side system via BGP FlowSpec [24]. While this avoids expensive hardware upgrades, it introduces significant communication overhead. More importantly, existing devices support only a limited number of filter rules [4, 34], making it difficult to handle large-scale botnet-generated attacks. Furthermore, under this strategy, the backbone side can only filter flows that have previously been identified by the access side. The access side must still perform fine-grained analysis on every newly incoming flow, which can overload the access side and ultimately limit the overall mitigation throughput.

NetSynergy employs a collaboration-based mitigation model approach to achieve backbone-side fine-grained DDoS mitigation. Using the characteristics of ongoing traffic and their access-side classification results, NetSynergy trains lightweight mitigation models in the access network. The model is designed with two key constraints: its input features must be extractable by the backbone side, and its inference complexity must remain within the backbone-side computational limits. These constraints ensure the model can be deployed on the backbone side without additional hardware. Furthermore, because the model targets only the current attack behavior, its resource requirements are significantly lower than those of complete access-side policies, making it suitable for the resource-limited backbone-side system. Once deployed, the backbone side can use the model to handle new flows exhibiting similar attack signatures, thereby scaling mitigation throughput. Overall, the collaboration-based mitigation model approach achieves a balance between deployment cost and mitigation performance, making it a more feasible and efficient design choice.

3.2 NetSynergy Architecture

NetSynergy builds the access-backbone collaborative defense using the collaboration-based mitigation model approach, with the decision tree serving as the mitigation model. Decision trees are selected for their advantages in line-rate dataplane inference, low training cost, and high interpretability. Firstly, the collaboration approach requires a machine learning model capable of executing directly on the dataplane to meet the high-throughput requirements of the backbone-side system. Prior works have proposed multiple line-rate decision tree inference schemes on programmable switches [18, 46, 47, 51, 55]. Secondly, decision trees can be trained rapidly, allowing the system to build mitigation models in real time and deploy timely DDoS defenses. Finally, their interpretability facilitates mitigation model tuning, allowing strategic adjustments on the backbone-side defense to balance throughput and accuracy. Based on these considerations, NetSynergy adopts a decision-tree-based mitigation model.

The architecture of NetSynergy is illustrated in Figure 2. The system takes traffic features and labels collected in the access network as input. The selected features are available on both access-side and backbone-side systems. Specifically, NetSynergy uses packet header features (e.g., `ip_protocol`, `ip_total_len`, `tcp_flags`), ensuring that the resulting mitigation models are backbone-deployable. These features are available in most scenarios, including defense against encrypted attacks. The corresponding labels are collected from existing access-side DDoS mitigation capabilities, which serve as external label providers (their internal design is outside the scope of this work). NetSynergy aligns features and labels using five-tuple information and maintains a continuously growing training buffer of aligned feature-label pairs. Using this dataset, NetSynergy trains mitigation models in the access network. The trained model is then deployed in the backbone network, where the Mitigation Model Inference mechanism performs line-rate DDoS mitigation. This mechanism is implemented based on NetBeacon [55], a line-rate decision tree inference scheme for programmable switches.

NetSynergy consists of two core components: Mitigation Model Generation in the access network (§4) and Traffic Dynamics Adaptation in the backbone network (§5). The former runs an iterative mitigation model training loop and governs the overall mitigation throughput and accuracy of the defense system during an attack. It

includes a Legitimate-Traffic-Preserving Model Generator, which minimizes misclassification of legitimate traffic (false positives) during training, and an Adaptive Model Confidence Controller, which monitors defense performance based on access-side throughput and dynamically adjusts the backbone-side mitigation aggressiveness accordingly in real time. The Traffic Dynamics Adaptation component operates primarily on the backbone-side dataplane. It implements a Model Update Fast Path mechanism that accelerates mitigation model convergence and adaptation to new attack patterns by prioritizing the most representative traffic samples for model updates. Specifically, these high-value traffic samples are selected directly on the dataplane with the Active Traffic Sample Selection mechanism. As NetSynergy operates, the mitigation model improves iteratively, eventually achieving high-throughput and accurate DDoS mitigation via access-backbone collaboration.

4 Mitigation Model Generation

Achieving high-throughput mitigation necessitates filtering the majority of malicious traffic on the backbone side. However, limited fine-grained backbone-side traffic analysis capabilities present a throughput-accuracy trade-off. Firstly, we develop a decision tree model training strategy that better preserves legitimate traffic during backbone-side mitigation. Secondly, to maintain sufficient overall mitigation throughput, we use an adaptive model confidence controller to dynamically adjust backbone-side mitigation throughput based on current defense performance.

4.1 Legitimate-Traffic-Preserving Mitigation Model Generator

In an access-backbone collaborative mitigation system, minimizing false positives (i.e., misclassifying legitimate traffic as malicious) in backbone-side mitigation is crucial for overall accuracy. Traffic destined for the protected network is first processed by the backbone-side system before being forwarded to the access network. Given that the access-side mitigation system offers more accurate traffic classification, the backbone-side system should filter only traffic exhibiting clear malicious signatures, while forwarding all other traffic (including suspicious flows lacking definitive evidence) for finer-grained access-side analysis. We term this strategy the Legitimate-Traffic-Preserving (LTP) mitigation principle, which ensures high overall mitigation accuracy of the access-backbone collaborative defense system.

To implement LTP within a decision tree, we apply two key adaptations that modify tree nodes prone to causing false positives: leaf node relabeling and tree structure refinement. Unlike the conventional approach that assigns leaf nodes to the majority class, we label leaf nodes based on the presence of legitimate training samples. Leaf nodes that contain legitimate training samples are labeled as benign, even if malicious samples outnumber legitimate ones within that node. During inference, traffic reaching such a leaf is conservatively forwarded to the access network for further analysis. We further refine the tree structure to reduce false positives. Standard decision tree training methods (such as ID3 or CART) aim to maximize class separation during tree construction. The LTP principle imposes an additional constraint that the feature

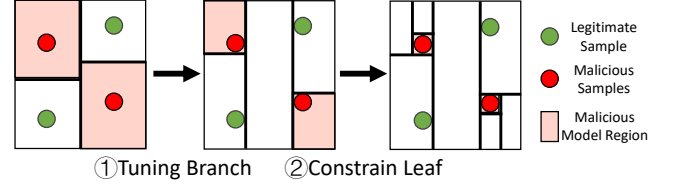


Figure 3: Legitimate-Traffic-Preserving Decision Tree.

region labeled as malicious must be minimized to reduce false positives. Accordingly, we modify the standard decision tree to build an LTP mitigation model. Specifically, we propose the following two structural modifications:

Tuning Branch Condition. We adjust existing branch conditions in the standard decision tree. Since the training dataset in NetSynergy is collected during an ongoing attack, early feature values often exhibit limited coverage of the feature space. The value sparsity can lead to unreliable threshold estimates, causing the model to misclassify benign traffic as malicious. Therefore, we tune the existing branch conditions to avoid classifying potential legitimate traffic as malicious based on the feature value range of known malicious samples (Step 1 in Figure 3). For each branch node along the decision path to a malicious leaf node, we analyze the range of malicious sample feature values in its child nodes. The branch threshold is then modified to match exactly the border of the known malicious feature region. If a branch condition lies on the decision path to two malicious leaves, we add an extra branch on the same feature to further separate them, ensuring that only traffic patterns closely matching known malicious behavior are classified as malicious.

Constraining Leaf Node Range. In addition to adjusting the existing branch conditions, we further restrict the feature region of leaf nodes to minimize misclassifications. Standard decision tree training selects only the path most beneficial for class separation, so the decision path may not necessarily include all the features. To address this, for each leaf node that classifies traffic as malicious, we introduce extra branch conditions based on the feature values of the malicious samples that reached that node during training (Step 2 in Figure 3). These boundaries are computed as the minimum and maximum feature values among those samples. We add an upper bound and a lower bound branch condition for each feature, thereby enabling the use of all features to avoid false positives. Note that legitimate leaf nodes do not undergo this processing, and if a particular branch condition already exists on the leaf's decision path, it is skipped in this process.

With these two modifications, we obtain a more robust decision tree model that significantly reduces the likelihood of legitimate traffic being misclassified. By tuning existing branch condition thresholds and introducing new branch conditions, we tighten the boundaries of the block behavior in the most conservative manner: only when incoming traffic patterns completely match known malicious samples in the dataset will they be blocked. However, such modifications introduce numerous extra branch conditions into the standard decision tree model, increasing the deployment cost on the dataplane. Therefore, the model must be pruned before being deployed on programmable switches.

Pruning Model for Dataplane Deployment. We apply a pruning strategy that removes redundant nodes and branches that do not affect the final forward / block decision. In an LTP decision tree, leaf nodes that are not confidently malicious are treated the same as benign ones during model inference. Therefore, multiple leaf nodes within the same subtree may lead to the same action when deployed on the dataplane. We identify such redundant leaf nodes and their corresponding branch conditions, then consolidate them into a single tree node. We iteratively remove such nodes until all remaining tree nodes are essential for separating forward and block actions in the LTP mitigation model. By removing redundant nodes, we reduce dataplane resource consumption without sacrificing accuracy or the LTP objective. Note that this model pruning process is performed before the tree structure modifications, as it reduces the number of tree nodes and thus decreases the computational overhead when calculating feature value ranges for each leaf node during the above modification processes.

4.2 Adaptive Model Confidence Controller

An access-backbone collaborative mitigation system should actively control the backbone-side system's mitigation aggressiveness to maintain sufficient overall mitigation throughput. Ideally, the majority of attack traffic is filtered by the backbone-side system, while only a small volume of traffic requiring further analysis is forwarded to the access-side system. This achieves an optimal balance between overall mitigation throughput and accuracy. However, various practical factors can disrupt this equilibrium. For example, the mitigation model may not have fully converged during initial system startup, the access-side black-box detector may provide incorrect traffic labels, or even worse, attackers might employ advanced attack strategies that prevent the system from training an effective mitigation model. In such cases, strictly following the LTP principle in §4.1 could lead to insufficient mitigation throughput. As the backbone-side mitigation model lacks the confidence to filter attack traffic accurately, it consequently forwards more traffic than the access-side system can process, leading to defense failure. To address this, we introduce an adaptive mechanism that dynamically adjusts backbone-side mitigation throughput, ensuring NetSynergy's robustness and protecting legitimate services even when mitigation model accuracy is suboptimal.

NetSynergy controls the backbone-side mitigation throughput by adjusting the target model confidence. Under the default LTP principle, the block action is performed only when the confidence level of a malicious-dominated decision tree leaf node reaches 100%. Reducing the target model confidence relaxes this threshold, allowing a leaf node to be classified as malicious if the proportion of training samples labeled as malicious exceeds the lowered confidence value. This adjustment enables the backbone-side system to filter a larger volume of traffic, thereby increasing its mitigation throughput to protect legitimate services. However, lowering the target confidence inherently involves a trade-off: while it boosts mitigation throughput, it also sacrifices mitigation accuracy, increasing the risk of legitimate traffic being incorrectly filtered. To balance mitigation throughput and accuracy, we adaptively adjust this parameter according to real-time defense status.

Algorithm 1: Target Model Confidence Control

Input: $Throughput_{T_i}$, $Throughput_{Target}$, $Delay_Window$

Output: $Target_Confidence_{T_i}$

```

1 STATE ← "STEADY"; // initial STATE
2 Confidence ← 100%; // initial confidence
3 Delay_Cnt = 0 // consecutive status count
  /* Execute once per decision interval */
4 switch STATE do
5   case "STEADY" do
6     if  $Throughput_{T_i} < Throughput_{Target}$ 
7       Delay_Cnt += 1;
8       if Delay_Cnt ≥ Delay_Window
9         Delay_Cnt = 0;
10        STATE ← "INCREASE";
11   else
12     Delay_Cnt = 0;
13     if Delay_Cnt ≤ -Delay_Window
14       Delay_Cnt = 0;
15       STATE ← "DECREASE";
16   case "DECREASE" do
17     if  $Throughput_{T_i} ≥ Throughput_{Target}$ 
18       Confidence = Step_Size;
19     else
20       STATE ← "STEADY";
21   case "INCREASE" do
22     if  $Throughput_{T_i} < Throughput_{Target}$ 
23       Delay_Cnt += 1;
24       if Delay_Cnt ≥ Delay_Window
25         Confidence = Step_Size;
26         Delay_Cnt = 0;
27     else
28       Confidence = Step_Size;
29       Delay_Cnt = 0;
30       STATE ← "STEADY";

```

The target model confidence control workflow is outlined in Algorithm 1. The system periodically monitors the traffic throughput at the access network $Throughput_{T_i}$ at each decision interval T_i and compares it with a predefined target throughput $Throughput_{Target}$. In the "STEADY" state, if $Throughput_{T_i} ≥ Throughput_{Target}$ for a consecutive number of intervals defined by $Delay_Window$, the system changes its state to "DECREASE" and begins lowering the target model confidence. This increases backbone-side mitigation aggressiveness, enabling the suppression of high-volume malicious traffic. Conversely, if $Throughput_{T_i} < Throughput_{Target}$ for $Delay_Window$ consecutive intervals, the system enters the "INCREASE" state and starts to increase the target model confidence, thereby forwarding more traffic to the access-side system for better mitigation accuracy. We adopt different strategies for "DECREASE" and "INCREASE" states. Target model confidence can be lowered aggressively in every interval, whereas increases are applied conservatively only once every $Delay_Window$ consecutive intervals. This design prioritizes maintaining sufficient mitigation throughput, as basic network availability is more important than

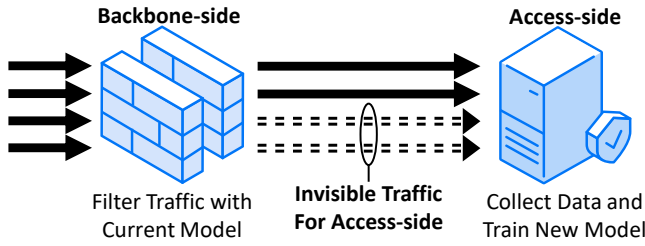


Figure 4: Partial Traffic Visibility in Access Network

accurate attack mitigation. Once the system stabilizes and can reliably defend against attacks, the target confidence may be cautiously increased to explore higher accuracy without compromising the required mitigation throughput. Through this adaptive mechanism, the backbone-side mitigation model dynamically balances accuracy and throughput, enabling robust DDoS mitigation.

5 Traffic Dynamics Adaptation

During a DDoS attack, adversarial strategies often evolve continuously, with new attack vectors emerging frequently. Therefore, NetSynergy must adapt to the traffic dynamics in real time, which is a particularly challenging task given that the fine-grained traffic analysis is performed in the access network and its processing throughput is constrained. In this section, we first formulate NetSynergy’s adaptation challenge as a special case of an active learning problem and propose a model update fast path mechanism within the collaborative mitigation architecture. We then design two high-value traffic sample selection algorithms deployable in the backbone-side system. These algorithms enable rapid discovery of new attacks and accelerate model convergence.

5.1 Model Update Fast Path

Model Iteration as Active Learning. NetSynergy’s mitigation model generation loop closely aligns with the active learning paradigm. The key hypothesis of active learning is that a model can proactively select its own training data. In typical active learning settings, training data is initially unlabeled, and the system iteratively selects a subset of samples to query an annotator for labels. The objective of the system is to maximize model performance while minimizing the number of labeling queries, making it particularly suitable for scenarios where labels are scarce or costly to acquire. In NetSynergy, unlabeled traffic samples continuously arrive, and the mitigation model is updated iteratively using these traffic samples. The access-side black-box detector, which acts as the annotator, provides corresponding traffic labels. As the detector runs at a limited processing throughput, the defense system must decide whether to request a label for each incoming traffic sample, based on both the detector load and the expected improvement to the mitigation model. From these perspectives, the access-backbone collaborative defense system can be interpreted as an active learning system. More specifically, NetSynergy’s workflow corresponds most closely to the stream-based selective sampling scenario [35].

However, the access-backbone collaborative defense scenario introduces a unique constraint: the unlabeled sample selection process must be deployed on the backbone-side system. In traditional

active learning, the system typically has access to all unlabeled samples and can freely select any of them for annotation. However, as Figure 4 shows, the backbone-side system filters part of the traffic before it reaches the access network. Because both traffic features and labels are extracted in the access network, the current model deployed at the backbone-side system constrains NetSynergy’s visibility into unlabeled samples. This constraint forces us to perform unlabeled sample selection before the backbone-side model filters the traffic. By performing selection upstream, the system maintains full visibility into all incoming traffic, allowing it to identify informative or representative samples that are valuable for mitigation model improvement. Such comprehensive visibility is critical, as traffic samples that contain new attacks or legitimate service patterns could otherwise be filtered and never contribute to model adaptation. Therefore, we implement the traffic sample selection in the dataplane and integrate it with a model update fast path mechanism.

Workflow. NetSynergy implements a straightforward model update fast path mechanism by prioritizing selected high-value traffic samples for access-side mitigation model generation. Active traffic sample selection algorithms deployed in the programmable switch dataplane identify samples that are most valuable for model improvement. Then, the relevant packets are duplicated on the dataplane. The original packets are processed normally (i.e., forwarded or dropped), while their copies are marked for priority processing and then forwarded to the access network. Marking is implemented by modifying header fields that are unused by both the access-side black-box detector and the backbone-side mitigation model. When these marked packets arrive at the access network alongside regular traffic, a strict priority queue is deployed to ensure they are processed first for feature extraction and detector analysis. Marked traffic is directed to the priority queue, while the unmarked traffic is placed in a regular queue. The access-side system prioritizes these packets, and the resulting features and labels are immediately incorporated into the training dataset for the next mitigation model generation cycle. By quickly integrating samples that were selected with full traffic visibility into the training pipeline, the system accelerates model performance improvement using the most informative and representative parts of the incoming traffic.

5.2 Active Traffic Sample Selection

Traffic sample selection must be performed before any traffic is filtered in the backbone-side system. To meet this requirement, NetSynergy designs traffic sample selection strategies deployable on the programmable switch dataplane. This enables line-rate selection of incoming traffic, and these strategies can be implemented on the same dataplane where the mitigation model inference runs. Specifically, NetSynergy implements two traffic sample selection strategies. The high model uncertainty strategy targets samples that are most valuable for improving model performance, based on an analysis of the decision tree structure. The unstable traffic behavior strategy focuses on discovering new attack traffic by identifying flows with anomalous and evolving characteristics.

High Model Uncertainty Strategy. The core principle of the high model uncertainty strategy is to prioritize traffic samples with high model uncertainty to accelerate model performance improvement.

In NetSynergy, both traffic classified as legitimate and traffic with undetermined status (LTP principle) are forwarded to the access network. During system initialization, the model has not yet converged, resulting in a large volume of unfiltered traffic that exceeds the processing capacity of the access-side black-box detector. Therefore, it is necessary to prioritize samples that contribute most to accelerating model convergence. The most straightforward approach is to focus on samples where model uncertainty is highest. Labeling these samples helps increase the proportion of traffic filtered by the updated model. For decision tree models, this means identifying traffic samples whose feature values lie near the decision boundaries that separate legitimate and malicious leaf nodes.

Specifically, the high model uncertainty strategy can be integrated into the mitigation model generation process in §4.1. This integration is achieved by introducing a special empty tree node for each additional branch condition added during model generation. These branch conditions tighten the feature range of malicious tree nodes, thereby precisely separating deterministic feature regions from uncertain ones. For example, when an original branch condition is further refined by adding an extra branch condition (Tuning Branch Condition in §4.1), an empty tree node is inserted to represent the uncertain region between the refined branches. Traffic samples matching these empty nodes are considered high-value candidates for model updates, as they belong to the uncertain region of the decision tree. When implementing this strategy on the dataplane, we can integrate it with the existing mitigation model inference. The classification result is extended to include an additional “Uncertain” alongside “Legitimate” and “Malicious”. When a mitigation model with this model uncertainty extension is transformed into table entries, the special empty tree nodes are mapped to the “Uncertain”, which subsequently triggers the model update fast path.

Unstable Traffic Behavior Strategy. The unstable traffic behavior strategy focuses on the discovery of unknown traffic patterns to enable rapid model adaptation. As mitigation models in NetSynergy are trained on known traffic samples, we cannot rely on existing models’ confidence estimates to find unknown attack behavior. Instead, this strategy monitors changes in traffic features to identify previously unseen attacks, leveraging the behavioral consistency within individual flows. Legitimate traffic typically exhibits stable feature patterns, whereas attacks often introduce abrupt and significant changes. By continuously tracking flow behavior and comparing it against its historical profiles, NetSynergy can identify behavioral changes and potential new attack strategies.

Specifically, NetSynergy implements lightweight traffic behavior monitoring on the dataplane. An ideal traffic change analysis would require fine-grained tracking of historical traffic characteristics (e.g., storing complete feature vectors per flow). However, such an approach is prohibitively expensive. To reduce overhead, we co-design the unstable traffic behavior monitoring with the model inference mechanism. Rather than storing raw historical features, NetSynergy maps each packet’s feature vector to a single integer representing the mitigation model’s classification result. For each monitoring window, we count the number of different classification results per flow and track their distribution. The behavioral characteristics of two consecutive windows are thus compressed

into two integers. Significant changes in these integer values indicate notable shifts in flow behavior, suggesting a potentially novel attack. When the difference between two consecutive windows exceeds a predefined threshold, subsequent packets from that flow are marked as “Uncertain” for a period of time, which triggers the model update fast path.

6 Implementation

NetSynergy Implementation. We implement NetSynergy’s backbone-side system on a Barefoot Tofino switch [16] and the access-side system on a commodity server. The dataplane decision tree inference mechanism is built on the open-source NetBeacon [55], extended with NetSynergy’s active traffic sample selection and model update fast path mechanisms. For the access-side system, traffic feature extraction is implemented with DPDK. These features are then labeled using outputs from the access-side black-box detector. The main workflow of NetSynergy’s mitigation model generation is implemented in Python on the access-side system. The generated model (and corresponding model boundary information for sample selection) is transmitted to the backbone-side system via HTTP. The access-side black-box detector is implemented on a separate commodity server. The detector outputs traffic labels and reports them to NetSynergy’s access-side system via HTTP. The detector also incorporates a priority-queue mechanism to prioritize the processing of high-value traffic samples.

Feature-Label Alignment. In NetSynergy, traffic features and labels must be aligned in a one-to-one correspondence before they can be used to train the mitigation model. However, due to differences in processing rates and latencies between the DPDK-based traffic feature extraction and the black-box detector, feature-label pairs often become misaligned as they arrive asynchronously. To manage this temporal misalignment, we introduce additional data structures to associate features and labels with their 5-tuple flow identifiers (i.e., source IP, destination IP, source port, destination port, and protocol). This approach necessitates efficient query and update operations over a large cache of unmatched features and labels. We implement two cache management optimizations: feature storage merging and LRU-based label management. Firstly, when identical traffic features appear repeatedly, we increment a counter for this feature instead of allocating additional storage; the accumulated count is then used to adjust the sample weight during model training. Secondly, as the detector continuously outputs new labels, the label set would otherwise expand indefinitely, degrading system performance. We maintain the label set using an LRU mechanism, which evicts the least-recently accessed label if the size reaches a predefined threshold.

Runtime Dataplane Model Update. NetSynergy iteratively updates the mitigation model to improve accuracy and adapt to evolving traffic patterns. This process requires runtime modifications to the dataplane model. To enable such updates, NetSynergy employs a runtime dataplane model update mechanism similar to Leo [18], which includes two decision tree inference structures and uses a flag register to indicate the currently active one. During each update cycle, the switch’s control plane receives a new decision tree model and translates it into corresponding table entries. These entries are then installed into the currently inactive inference structure

Table 1: DDoS Mitigation Performance of NetSynergy and Baselines.

Dataset	CIC-IDS-2018				CIC-DDoS-2019				BCCC-cPacket-Cloud-DDoS-2024			
Metric	BFR	FNR	B-FPR	CDR	BFR	FNR	B-FPR	CDR	BFR	FNR	B-FPR	CDR
No Defense	39.95%	39.00%	-	60.81%	26.37%	25.61%	-	73.70%	27.99%	26.51%	-	72.47%
Ideal Access	38.16%	0.00%	-	62.45%	23.78%	0.00%	-	75.35%	27.06%	0.00%	-	73.48%
NetBeacon	87.24%	2.76%	12.76%	0.00%	90.69%	0.65%	9.31%	0.00%	63.57%	24.71%	10.76%	13.01%
Planter	77.15%	9.25%	22.96%	0.00%	54.39%	7.85%	27.83%	9.97%	78.21%	7.04%	22.35%	0.00%
SISTAR	92.16%	1.31%	8.15%	0.00%	59.87%	3.02%	40.98%	0.00%	75.16%	13.66%	13.18%	2.18%
ACL-Coord	75.33%	0.00%	0.00%	17.19%	41.18%	0.00%	0.00%	55.89%	43.31%	0.00%	0.00%	44.88%
NetSynergy	95.07%	0.00%	0.62%	4.40%	91.47%	0.00%	2.11%	7.39%	89.33%	0.00%	5.35%	5.14%

on the dataplane. Once the installation completes, the flag register state is modified to activate the new decision tree and disable the old one. This design eliminates the need for dataplane program recompilation and thus incurs no switch downtime during updates. Moreover, it ensures atomic model switching, preventing errors that could arise from packets accessing partially updated states.

7 Evaluation

In this section, we evaluate NetSynergy from three aspects: (1) End-to-end performance in mitigating high-volume DDoS attacks. NetSynergy provides more effective mitigation and better protection for legitimate services compared with state-of-the-art solutions; (2) Mitigation performance under dynamic attack scenarios. NetSynergy achieves robust DDoS mitigation when attack patterns change dynamically; (3) Design validation. We further test the performance gain of individual mechanisms, mitigation throughput, and the deployment overhead of NetSynergy.

7.1 Experiment Setup

Testbed Setup. Based on the NetSynergy prototype described in §6, we set up an access-backbone collaborative defense testbed. The test traffic first passes through a programmable switch, which acts as the backbone-side mitigation system, before being forwarded to a server hosting the access-side mitigation system. To emulate a bandwidth-constrained access-backbone link, we apply a fixed-rate limiter that restricts available bandwidth to half of the test traffic throughput at the switch. Packet drops resulting from this rate limiting are logged at the switch. Besides, each packet in the test packet trace is tagged with a ground truth label (used only for monitoring, not accessible to any mitigation scheme). Both the switch and the server record per-packet decisions based on these tags, allowing for precise collection of performance metrics.

Baselines. We compare NetSynergy with existing DDoS mitigation and dataplane traffic classification schemes, including:

- No Defense: A baseline with no DDoS mitigation deployed, representing the worst-case mitigation performance.
- Ideal Access: A baseline with ideal access-side DDoS mitigation, representing the best-case mitigation performance with no backbone-side assistance.
- NetBeacon [55]: A dataplane traffic classification scheme using a decision tree model. We use only the packet-level features and the model in the open-source NetBeacon.

Table 2: Detailed Mitigation Performance of NetSynergy

Dataset	B-TPR	A-TPR
CIC-IDS-2018	85.80%	9.78%
CIC-DDoS-2019	90.13%	1.94%
BCCC-cPacket-Cloud-DDoS-2024	89.09%	5.83%

- Planter [52]: A dataplane machine learning inference scheme. We adopt the XGBoost (XGB) model (the best performance configuration in [52]) and the feature set from their CICIDS scenario. We extend Planter’s dataplane with the missing traffic feature extraction functionality.
- SISTAR [14]: A dataplane DDoS detection and mitigation scheme using an improved decision tree model for lower resource consumption. We use the feature set from the open-source SISTAR and follow its decision tree training process. We modify the original SISTAR on BMv2 [29] to build a version on TNA [3] that runs in our testbed.
- ACL-Coord: A naive approach to coordinating access-side and backbone-side mitigation systems. The access side detects malicious traffic flows and converts their flow IDs into ACL rules, which are then deployed on the backbone-side system. IP pairs are used as the keys for ACL entries.
- NetSynergy: The prototype as described in §6. We set the parameter $Throughput_{Target}$ in Algorithm 1 slightly above the peak rate of benign traffic throughput (which NetSynergy’s user can estimate from historical peak values). NetSynergy’s mitigation model is trained from scratch.

Note that machine-learning-based baselines are all trained offline on the entire test traffic, with their models remaining unchanged during the experiments. This configuration gives the baselines an unfair advantage and is impractical for real-world deployment, since future traffic is inherently unpredictable. In addition, due to limited dataplane storage space, we modify the flow definitions of Planter and SISTAR to use source IP as flow ID, given the large number of flows in our scenario.

Test Traffic. We construct high-volume DDoS traffic using publicly available datasets. We use traffic from MAWI [45] as the benign traffic of the protected victim services, while attack traffic is from CIC-IDS-2018 (HULK, LOIC, HOIC, Goldeneye), CIC-DDoS-2019 (LDAP, MSSQL, NetBIOS, Portmap, Syn, UDPLag, UDP) [37], and

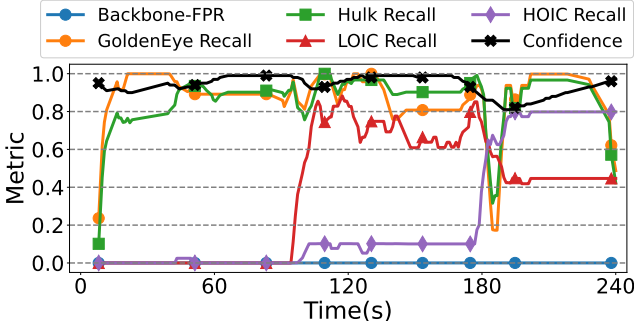


Figure 5: Model Performance under Dynamic Attack

BCCC-cPacket-Cloud-DDoS-2024 (17 TCP-based Attacks) [36]. We process the traffic dataset to ensure that the ratio of malicious to benign samples and the total traffic volume are consistent across the three attack datasets.

Metrics. We evaluate DDoS mitigation performance using four metrics: **Benign Traffic Forwarding Rate (BFR)**, **False Negative Rate (FNR)**, **Backbone-side False Positive Rate (B-FPR)**, and **Congestion Drop Rate (CDR)**. Firstly, BFR and FNR capture the overall effectiveness of DDoS mitigation. BFR is the proportion of legitimate packets successfully delivered to the victim server. FNR is the proportion of malicious packets that reach the victim server. BFR reflects the availability of legitimate services under attack, while FNR measures the amount of attack traffic that bypasses all mitigation systems. Secondly, we further report the detailed mitigation performance of the backbone-side systems using the B-FPR and CDR. B-FPR is defined as the ratio of incorrectly blocked legitimate packets to total legitimate packets, reflecting misclassification of benign traffic. CDR is defined as the ratio of packets dropped due to link congestion during an attack to total packets; a higher CDR indicates that the scheme fails to mitigate a sufficient volume of malicious traffic. In this setting, a higher BFR is preferred, while lower values of the other three metrics are preferred.

7.2 End-to-End Performance

In this section, we compare NetSynergy with existing DDoS mitigation systems using the test traffic described in §7.1. Results are shown in Table 1. We also report the volume of traffic filtered at each side (access and backbone) of NetSynergy in Table 2.

Our evaluation constructs a DDoS attack scenario where the traffic volume exceeds the capacity of the access-backbone link. The No Defense baseline shows that the attack causes significant loss of legitimate traffic. In this situation, the performance of access-side-only mitigation solutions is extremely limited. Therefore, although the Ideal Access can completely eliminate attack traffic reaching the server (0% FNR), the benign traffic loss that occurs in the congested link cannot be avoided (low BFR score).

Dataplane baselines (NetBeacon, Planter, SISTAR, ACL-Coord) are deployed on the backbone side. They leverage the backbone's bandwidth advantage to achieve sufficient mitigation throughput, thereby avoiding the negative impact of packet loss on the access-backbone link. However, NetBeacon uses a standard decision tree model training method where leaf nodes are assigned the majority

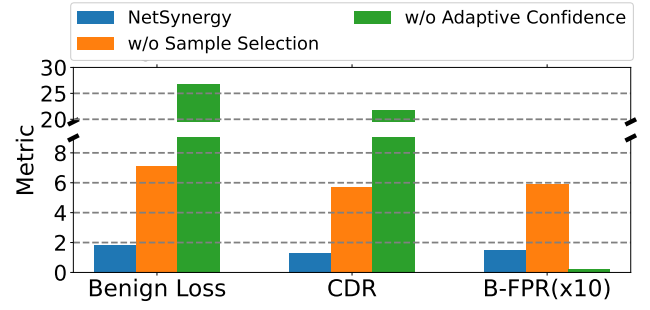


Figure 6: Ablation Test on NetSynergy

class, leading to more misclassifications of benign traffic compared to NetSynergy's LTP model. For flow-level traffic analysis schemes (Planter and SISTAR), we observe that they are prone to performance degradation due to insufficient concurrency when facing DDoS attacks. Although these baselines are all trained offline using the full dataset, they experience significant performance losses in the testbed compared to the training phase. This is because the number of flows in our test scenario (identified by source address) is comparable to the number of flow entries supported by Tofino's storage, and potential hash collisions in the flow table adversely affect subsequent model inference. In fact, this is why we modified the flow definitions of these two baselines; the original 5-tuple granularity performed even worse. ACL-Coord uses an access-side blacklist to block attacks, achieving a high defense accuracy with zero false positives. However, due to the large number of attack flows, the overhead of transmitting malicious flow IDs and inserting them into the dataplane table is significant. In addition, the latency and non-generalization issues analyzed in §3 also make its performance less than ideal.

NetSynergy achieves the best overall mitigation performance across all test settings through access-backbone collaboration. It achieves the highest benign traffic forwarding rate while maintaining the lowest false negative rate, offering the strongest protection for legitimate services. Compared with Planter, NetSynergy achieves 37.08% improvement in BFR in the CIC-DDoS-2019 test. Detailed results in Table 2 show that more than 85% of malicious traffic is filtered on the backbone side in all tests, which means most attack traffic is filtered upstream, even considering the misses that occur during the initial system warm-up period. NetSynergy quickly converges to a stable state where both high mitigation throughput and legitimate traffic preservation are achieved, providing robust service protection. Note that the slight performance loss in B-FPR and CDR is mainly due to the system startup process. Because NetSynergy generates models online from scratch rather than deploying a static pre-trained model, some traffic drops and false positives occur before an effective model is established.

7.3 Mitigating Dynamic Attack

In this section, we evaluate NetSynergy's performance under dynamic attack scenarios. We modify the CIC-IDS-2018 test traffic described in §7.1 so that four attacks (GoldenEye, Hulk, LOIC, and HOIC) are launched sequentially rather than simultaneously. We

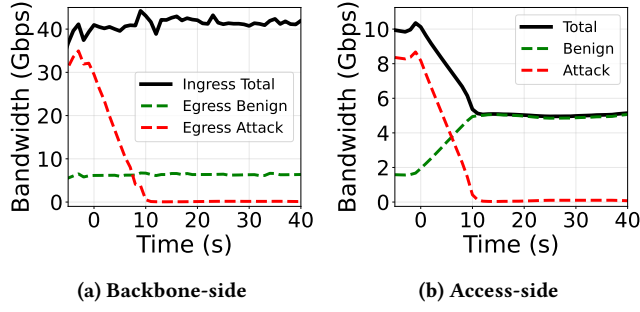


Figure 7: NetSynergy's Mitigation Throughput

record all mitigation models generated during the experiment along with their corresponding target model confidence. The mitigation performance of these models is then evaluated offline on the complete packet trace, and the key metrics are reported in Figure 5.

Under the dynamic attack scenario, NetSynergy achieves an overall benign traffic forwarding rate of 91.99%. From the perspective of real-time model performance, NetSynergy maintains a consistently low false positive rate for legitimate traffic while rapidly adapting to newly introduced attacks. The recall for new attack traffic quickly rises to a level that ensures accurate mitigation. During the attack transitions, the adaptive model confidence controller adjusts mitigation aggressiveness to maintain sufficient mitigation throughput: confidence decreases when mitigation temporarily fails and gradually recovers as the defense stabilizes, thereby dynamically balancing mitigation throughput and accuracy in response to the current defense status. Note that the reported model performance is based on offline evaluation, which differs from the tested performance because each mitigation model is only deployed for a short time. Although this introduces some evaluation bias, it does not significantly affect our analysis. We also monitor real-time access-received traffic throughput (before the access-side mitigation). The results show that the system maintains effective backbone-side mitigation with little attack traffic reaching the access network for most of the experiment, except for brief intervals during attack transitions where attack traffic volume increases. In summary, NetSynergy delivers robust and adaptive DDoS defense in the face of dynamically changing attacks.

7.4 Deep Dive

Ablation Test. To fully understand the performance gain of NetSynergy's designs on DDoS mitigation, we conduct an ablation study on NetSynergy. We build a NetSynergy without the Active Traffic Sample Selection mechanism (i.e., it passively collects training samples from access-received traffic) and a NetSynergy without the Adaptive Model Confidence Controller mechanism, which strictly follows the legitimate-traffic-preserving principle during the mitigation model generation. Three solutions are evaluated with CIC-IDS-2018 test traffic in §7.1. Results are shown in Figure 6.

The differences in Benign Loss (defined as $1 - \text{BFR}$) across three schemes indicate that removing either key mechanism from NetSynergy leads to notable performance degradation. The variant without the adaptive confidence mechanism suffers more performance loss, as it cannot adjust its defense strategy in response to the

Table 3: Access-Backbone Collaboration Gain Test

	BFR	B-TPR	B-FPR
NetSynergy (Tofino)	95.07%	85.80%	0.62%
NetSynergy (DPDK)	91.35%	84.75%	1.90%
NetBeacon[55]	87.24%	96.32%	12.76%
Jaquen[23]	46.13%	29.48%	12.85%

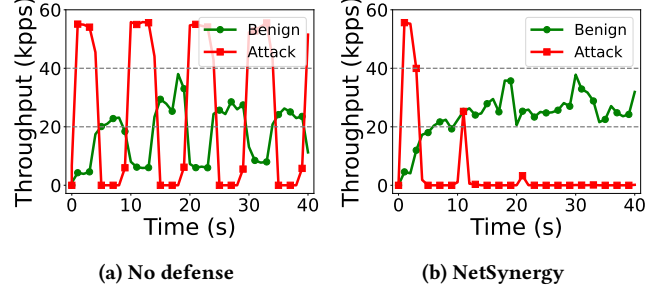


Figure 8: Pulse Attack Test

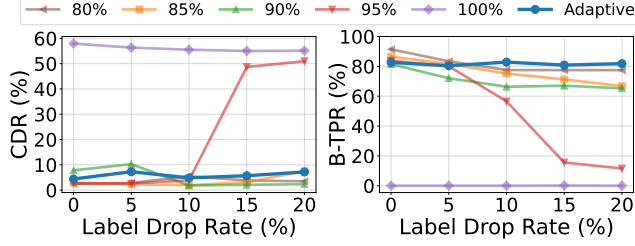
current mitigation performance. Its conservative model generation strategy limits backbone-side mitigation throughput, resulting in an extremely high CDR. Although this variant achieves the lowest backbone-side false positive rate, it clearly fails to protect legitimate services during DDoS attacks. On the other hand, removing active sample selection causes a relatively smaller performance drop. Without the ability to prioritize high-value traffic samples, this variant can only incorporate passively received traffic into the training set. This slows down mitigation model convergence, extending the time required to achieve sufficient mitigation throughput and accuracy. The extended unstable state during startup raises both CDR and B-FPR, ultimately degrading the overall mitigation performance.

Collaboration Gain Test. Beyond Access-Backbone Collaboration, NetSynergy also leverages programmable data planes and machine learning models to enhance defense. To isolate the contribution of collaboration, we devise a Gain Test. We reproduce a dataplane DDoS defense system, Jaquen [23], as a representative of schemes that use predefined defense policies such as Bohatei [11] and Poseidon [50]. We also implement NetBeacon [55], which uses the same decision-tree model as NetSynergy but disables collaboration. We additionally implement a software version of NetSynergy using a DPDK-based backbone-side system. All experiments use the CIC-IDS-2018 test traffic in §7.1, and results are shown in Table 3.

Both Tofino-based and DPDK-based NetSynergy outperform the non-collaborative schemes. The performance of Jaquen is representative of fixed-rule DDoS defense systems. These schemes offer high-throughput filtering, but due to their reliance on manually configured defense policies, they fall short of ML-based methods in terms of mitigation accuracy. The DPDK version still surpasses non-collaborative baselines despite lower processing capacity than Tofino, highlighting collaboration's value. Performance gaps among NetSynergy variants are mainly related to the mitigation model collaboration mechanism. Backbone-model replacements require rapid response of backbone-side system implementation for timely deployment, which influences the system's defense effectiveness

Table 4: Legitimate-Traffic-Preserving Model Generation

	FPR	Accuracy	Dataplane Cost
Standard DT	5.062%	98.89%	2.77KB
LTP model	0.508%	88.80%	7.74KB (raw 11KB)

**Figure 9: NetSynergy Robustness Testing**

during the warm-up phase. In our experiments, DPDK takes 1-2 seconds longer to establish defenses than Tofino, resulting in 9.35% CDR and worse BFR.

Mitigation Throughput. NetSynergy’s dataplane program does not incur any packet recirculation, theoretically supporting line-speed processing. We test NetSynergy’s mitigation throughput on our testbed, sending approximately 40Gbps of test traffic using XL710 network cards [15] and DPDK-pktgen [17]. We modify the CIC-IDS-2018 test traffic to a benign-to-malicious traffic ratio of 1:10. The throughput collected during the experiment is shown in Figure 7. We smooth the data using a 10-second average window.

NetSynergy successfully achieves 40Gbps DDoS defense, with a measured maximum malicious traffic blocking throughput of 34.96Gbps. The final unblocked malicious traffic volume is approximately 150Mbps, negligible compared to the benign traffic volume. From the time the first access-side label is received, NetSynergy takes approximately 5 seconds to establish an effective defense. This test demonstrates NetSynergy’s ability to handle high-volume DDoS mitigation.

Pulse Attack Test. Pulse attacks specifically challenge the response time of defense systems[2]. We modify the CIC-IDS-2018 test traffic to construct a pulse attack with a 5s attack and 5s idle intervals, as in ACC-Turbo [2]. The throughput collected at the access side is illustrated in Figure 8 for both the no defense scenario and the NetSynergy defense scenario (the experiment lasted 90s, with 40s shown in Figure 8 for brevity).

NetSynergy achieves effective defense against pulse attacks, raising BFR from 68.16% to 97.65%. Specifically, NetSynergy already delivers effective mitigation within the first pulse. Although a few malicious packets are not filtered in the second and third pulses, they do not affect the benign traffic forwarding, and subsequent pulse attacks are fully blocked. Excluding a B-FPR of 0.6%, only 1.75% of benign traffic is lost during warm-up. NetSynergy’s response time is sufficient for pulse-attack defense. Note that we also evaluated pulse attacks with shorter attack and idle intervals (ranging from 1s to 5s). NetSynergy maintains effective defense across all tested configurations, achieving a BFR exceeding 90%. For brevity, we only report detailed results for one pulse attack configuration.

Table 5: Resources Usage in Programmable Switch

Resources	Percent
Meter ALU	8.3%
Hash Bit	2.4%
SRAM	6.3%
TCAM	19.8%

Legitimate-Traffic-Preserving Model Generation. NetSynergy generates mitigation models under the LTP principle, which prioritizes the protection of legitimate services during backbone-side mitigation. To validate the effectiveness of this design, we build a standard decision tree model and an LTP model offline, using the CIC-IDS-2018 test traffic in §7.1. We use a target confidence of 99% in LTP model generation. Results are shown in Table 4.

Compared with the standard decision tree, the LTP model significantly reduces the false positive rate, thereby introducing fewer misclassifications on legitimate traffic in backbone-side mitigation. Although this improvement comes at the cost of a slight decrease in classification accuracy, such a trade-off is acceptable in access-backbone DDoS mitigation scenarios, as any malicious traffic missed by this model can still be filtered by the access-side system. The model pruning mechanism reduces the LTP model size to within acceptable limits for deployment on programmable switches, achieving nearly a 30% reduction compared with the original LTP decision tree. Note that the offline model performance reported here cannot be directly compared with testbed results in Table 1.

NetSynergy Robustness Testing. NetSynergy relies on the access-side detection results as system input; therefore, its defense performance, model accuracy, and convergence speed are all influenced by the access-side detector. To enhance robustness, we introduce adaptive model confidence in §4.2. To evaluate this design, we add random label drops to the ideal access-side detector, simulating degraded access-side detector performance. We compare the original NetSynergy against variants with static model confidence configurations. Results are shown in Figure 9.

The variant strictly following the LTP principle (labeled 100%) fails to achieve sufficient backbone-side defense throughput (B-TPR) across all tested scenarios, resulting in significant packet loss (CDR). This indicates that overly strict confidence requirements undermine robustness, as inaccuracies in any access-side label can substantially degrade performance. Our experiments further show that moderately lowering confidence improves resilience to access-side instability: a static confidence below 90% can achieve sufficiently high defense throughput. NetSynergy with adaptive confidence achieves robust DDoS mitigation under all label drop rates. NetSynergy adaptively adjusts model confidence based on real-time mitigation performance, balancing throughput and accuracy. This also allows NetSynergy to achieve higher throughput than static low-confidence variants, particularly in high-drop-rate scenarios.

Access-Backbone Transmission Cost. NetSynergy adapts to traffic dynamics through the Model Update Fast Path mechanism in §5.1. High-value traffic samples, selected by the backbone-side system, are duplicated and transmitted to the access-side system. This approach introduces additional packet transmission. To quantify the cost, we record the number of marked packets (duplicated

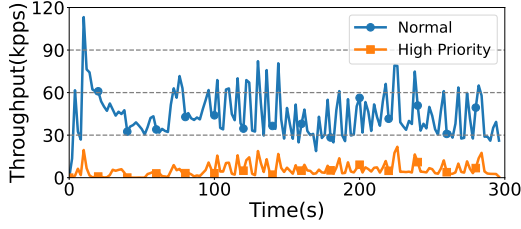


Figure 10: Prioritized Samples Transmission Cost

high-priority samples) and normal packets (original traffic) during the end-to-end performance test. Results are shown in Figure 10.

The duplicated traffic throughput remains at a low level throughout the experiment, with an average throughput of 5.36kpps. It constitutes only a small fraction of the total access-received traffic. Given the significant advantages of this mechanism in improving system mitigation performance (see the Ablation Test), we consider the overhead acceptable. Additionally, there are some potential methods for further reducing the transmission cost (discussed in §8). Note that transmitting mitigation models from the access side to the backbone side also incurs transmission cost. However, since NetSynergy updates the model at one-second intervals, the resulting transmission throughput is negligible.

Hardware Resource Consumption. Table 5 presents the hardware resource utilization of NetSynergy. The reported values are measured with a prototype using 6 features, each mapped to a 32-bit encoding, and all encoding tables (feature encoding and tree encoding) are configured to hold up to 1000 entries. This setup aligns with the configuration used in our evaluation and is adequate to achieve satisfactory mitigation performance. There are still sufficient resources available for deploying additional network functions or adopting more sophisticated models in NetSynergy.

8 Discussion

Prioritized Traffic Sample Transmission. NetSynergy transmits prioritized traffic samples by replicating and forwarding selected packets; however, this approach is far from perfect. Firstly, performing feature extraction on the backbone side could further reduce transmission overhead. Once the feature set required by the access-side and backbone-side mitigation is determined, a backbone-side feature extraction mechanism can be deployed to generate more compact representations of the prioritized traffic. However, its feasibility depends on the deployment cost of the backbone-side feature extraction mechanism. Secondly, in practical deployments, encapsulating the original traffic sample data should be considered. For instance, the original information can be embedded as the payload of ICMP, TCP, or tunneling protocols. This approach avoids modifying the original packet headers, which could otherwise cause intermediate network devices to mistakenly trigger man-in-the-middle attack warnings and block the transmission.

Dual-Label Detector Requirements. NetSynergy requires the access-side detector to output both legitimate and malicious traffic labels. The LTP model generator in NetSynergy adopts an aggressive strategy to protect legitimate samples (§4.1). Consequently, the system depends on both label types to maintain its mitigation

throughput. If the access-side detector only reports malicious labels (e.g., Snort [6]), NetSynergy must treat all unlabeled samples as legitimate. If there are unlabeled malicious samples in the training buffer, the LTP model’s performance would significantly degrade. Our experiments with a single-label detector confirm that such a configuration leads to poor system stability. Treating unlabeled traffic as legitimate prevents the mitigation model from filtering sufficient attack traffic. Therefore, although reporting only malicious traffic labels could significantly reduce the transmission overhead, NetSynergy adopts a more stable dual-label approach.

Limitation. NetSynergy’s effectiveness hinges on several prerequisites. First, reliable access-side labels must be available; attacks unrecognizable at the access side inherently fall outside the system’s mitigation scope. Second, attack patterns must be learnable given the constraints on the collaboration-based mitigation model approach (the features used for mitigation must be accessible on both sides, and the complexity of the mitigation model is constrained), potentially allowing adversaries to craft evasion attacks. However, both of these scenarios would force attackers to pay higher costs to craft complex attacks specifically against NetSynergy’s defenses. Third, when facing previously unseen attacks, NetSynergy requires a warm-up period before its defense mechanism becomes fully effective. During this interval, the attackers could continuously alter their strategies to keep the defense ineffective. Nevertheless, since NetSynergy performs online learning, persistent evasion would demand that every new attack be entirely distinct from all prior instances, which is difficult and substantially raises the attack cost.

9 Related Works

Multi-layer Traffic Classification. NetSynergy adopts a layered DDoS mitigation strategy: it filters the majority of attack traffic on the backbone-side system, while handling a small, more complex portion on the access-side system. This layered approach is widely used in real-time traffic classification. For instance, HorusEye [9] uses isolation forests on the dataplane to analyze high-volume traffic, forwarding only suspicious flows to the control plane for in-depth inspection. Similarly, BoS [49] partitions the traffic analysis workload by model confidence, with the majority of the traffic processed by binary RNNs on programmable switches and only a small portion analyzed with more advanced models. Unlike prior works, NetSynergy performs layered classification on physically separated systems (access and backbone). Furthermore, NetSynergy is designed with full awareness of this sequential mitigation scenario, and its mechanisms (e.g., the LTP model and dataplane active sample selection) are specifically tailored to it, enabling high-throughput and accurate DDoS mitigation.

Dataplane Model Inference. Performing machine learning model inference directly on the dataplane can significantly improve throughput and reduce latency, which is highly valuable for real-time applications such as DDoS defense, traffic classification, and intrusion detection. Several studies have explored the deployment of decision tree models on programmable switches [14, 18, 46, 47, 51, 52, 55]. Beyond decision trees, Taurus [39] proposes a dedicated chip architecture for executing neural network models at line rate, while BoS [49] leverages both programmable switches and commodity

servers equipped with GPUs to perform RNN inference. NetSynergy currently employs decision tree models with NetBeacon's tree embedding approach [55]; it can also readily adopt other decision-tree-based deployment schemes [14, 18, 46, 47, 51, 52]. Supporting other model types (e.g., neural networks) would require additional efforts in LTP model generation, which we leave for future work.

10 Conclusion

Application-layer attacks exploit a key weakness of existing DDoS defense systems: the restricted backbone-side mitigation capabilities. As a result, malicious traffic can easily bypass the backbone-side defense, saturating access network bandwidth and disrupting legitimate services. To address this challenge, we propose NetSynergy, an access-backbone collaborative DDoS mitigation system. NetSynergy enables accurate and high-throughput DDoS mitigation via a collaboration-based mitigation model approach. Evaluations on a hardware prototype demonstrate that NetSynergy outperforms existing solutions and provides better protection for legitimate services under high-volume application-layer attacks.

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A Open Science

In this section, we provide information about obtaining the source code and dataset of NetSynergy. We open-source NetSynergy prototype codes and dataset in Zenodo (<https://doi.org/10.5281/zenodo.19871383>). Researchers should be able to run NetSynergy on their own devices. Due to performance fluctuations in hardware devices, we cannot guarantee that all results will be completely consistent as in this paper. But the results should be similar and able to validate the innovations and advantages we claim.

NetSynergy prototype is implemented for hardware devices, and no simulation version is available. We provide hardware requirement list as follows:

- Backbone-side system: a programmable switch (e.g., Tofino-based switch) that supports P4 and rate-limiting functions.
- Access-side system: a server with DPDK support, connected to the backbone-side system. This server runs a DPDK process and a Python process for mitigation model generation.
- Access-side black-box detector: a server, connected to the backbone-side system. This server runs a detector process, a priority queue process and a HTTP client process for providing traffic labels and implementing traffic dynamics adaptation component.
- Traffic generator: a server, connected to the backbone-side system. This server replays the traffic traces to the backbone-side system.

B Ethical Considerations

This research aims to enhance the accuracy and throughput of DDoS mitigation techniques, with no ethical issues arising from the research process. The study targets known DDoS attack types and does not introduce any novel attack methods. The proposed system, NetSynergy, is designed to be compatible with existing DDoS defense solutions and does not disrupt existing security mechanisms.

All experiments in this paper are conducted within a controlled testbed environment, isolated from real network environments. NetSynergy is never deployed or tested in any real production networks or online services. The experimental dataset is sourced exclusively from publicly available datasets (MAVI [45], CIC-IDS-2018, CIC-DDoS-2019 [37], and BCCC-cPacket-Cloud-DDoS-2024 [36]). These datasets are only processed as necessary to fulfill experimental objectives, and thus do not involve any user-private information.